

SQL Server Management MCP Server - Complete Implementation and Usage Guide

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Note: This is a simplified HTML version for easy printing to PDF. For the complete markdown version, see `docs/COMPLETE_GUIDE.md`

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Executive Summary

The SQL Server Management MCP (Model Context Protocol) Server is a comprehensive solution for managing Microsoft SQL Server instances through natural language interactions with AI assistants like OpenCode.

Key Features

- **Windows Authentication:** Seamless integration with existing UCN credentials
- **Environment Protection:** Automatic prevention of write operations on production servers
- **Comprehensive Logging:** Full audit trail of all operations
- **Error Handling:** Robust error management with detailed feedback

Available Operations

- **Database Querying:** Read-only access to dbo schema tables
- **Data Modification:** Controlled write operations (DEV environments only: DOA, DOB)
- **Job Monitoring:** SQL Server Agent job status and history tracking
- **Instance Management:** Server status monitoring and service control

Prerequisites and Installation

System Requirements

Component	Requirement
Operating System	Windows Server 2016+ or Windows 10+
Python Version	3.9 or higher
Memory	2 GB RAM minimum
Disk Space	100 MB

Installation Steps

Quick Install:

```
cd C:\path\to\sql-server-mcp
.\install.ps1
```

1. Install ODBC Driver

1. Download Microsoft ODBC Driver 17 for SQL Server
2. Run the installer: `msodbcsql.msi`
3. Accept license terms and complete installation

2. Install Python Dependencies

```
cd C:\path\to\sql-server-mcp
pip install -e .
```

3. Configure OpenCode

Add to `~/.config/opencode/config.json`:

```
{
  "mcpServers": {
    "sql-server": {
      "command": "python",
      "args": ["-m", "src.server"],
      "cwd": "C:/path/to/sql-server-mcp",
      "env": {
        "PYTHONPATH": "C:/path/to/sql-server-mcp",
        "LOG_LEVEL": "INFO"
      }
    }
  }
}
```

Configuration

Environment Detection

The MCP server automatically detects the environment based on server naming:

Server Name Pattern	Detected Environment	Write Operations
Contains "DOA"	Development A (DOA)	✓ Allowed
Contains "DOB"	Development B (DOB)	✓ Allowed
Contains "PROD", "PRD", "PRODUCTION"	Production	✗ Blocked
Other	Unknown	✗ Blocked

Tool Reference

Table Operations (6 tools)

1. list_tables

List all tables in the dbo schema of a database.

Parameters:

- server_name: SQL Server instance name
- database_name: Target database

2. get_table_schema

Get detailed schema information for a table.

3. query_table

Execute SELECT queries on dbo tables.

4. update_table (DEV ONLY)

 **RESTRICTION:** Only works on DOA or DOB servers. WHERE clause is REQUIRED.

5. insert_into_table (DEV ONLY)

Insert new records into a table (DOA/DOB only).

6. delete_from_table (DEV ONLY)

Delete records from a table (DOA/DOB only).

SQL Agent Job Operations (4 tools)

7. list_agent_jobs

List all SQL Server Agent jobs.

8. **get_job_history**

Get execution history for SQL Server Agent jobs.

9. **get_job_status**

Get current status of a specific job.

10. **get_failed_jobs**

Get jobs that failed in the specified time window.

Instance Management Operations (5 tools)

11. **get_instance_status**

Get SQL Server instance status and information.

12. **restart_sql_service**

 **REQUIRES:** Administrator privileges

13. **restart_agent_service**

Restart the SQL Server Agent service.

14. **get_service_status**

Check Windows service status for SQL Server or Agent.

15. **test_connection**

Test connectivity and authentication to SQL Server.

Security and Access Control

Environment-Based Protection

- **Allowed:** DOA, DOB environments

- **Blocked:** Production and unknown environments
- **No Bypass:** Hard-coded protection cannot be disabled

Authentication

- Windows Authentication (UCN credentials)
- No passwords in configuration
- SQL Server role-based permissions
- Parameterized queries prevent SQL injection

Usage Examples

Example 1: Query Data

User: "Show me all users from DOA-SERVER\DEV database CustomerDB"

Tool: query_table

Result: Users displayed with all columns

Example 2: Update Dev Data

User: "Update user status to 'inactive' for user ID 101 in DOA database"

Tool: update_table

Environment: DOA (✓ allowed)

Result: 1 row updated

Example 3: Production Protection

User: "Update user status in PROD-SERVER"

Tool: update_table

Environment: PRODUCTION (X blocked)

Result: Error - write operations not allowed on production

Troubleshooting

Connection Timeout

Error: Unable to connect to server. Timeout expired.

Solutions:

- Verify server is online
- Check network connectivity
- Verify firewall settings
- Check server name spelling

Authentication Failed

Error: Login failed for user 'DOMAIN\username'

Solutions:

- Grant SQL Server login permissions
- Verify Windows Authentication is enabled
- Check domain and username

Permission Denied on Write Operations

Error: UPDATE operations are only allowed on DEV environments

This is by design! Write operations only work on DOA and DOB servers.

Additional Resources

For complete documentation, see:

- README.md - Quick start guide
- docs/COMPLETE_GUIDE.md - Full documentation (1,500+ lines)
- PROJECT_SUMMARY.md - Project overview

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